

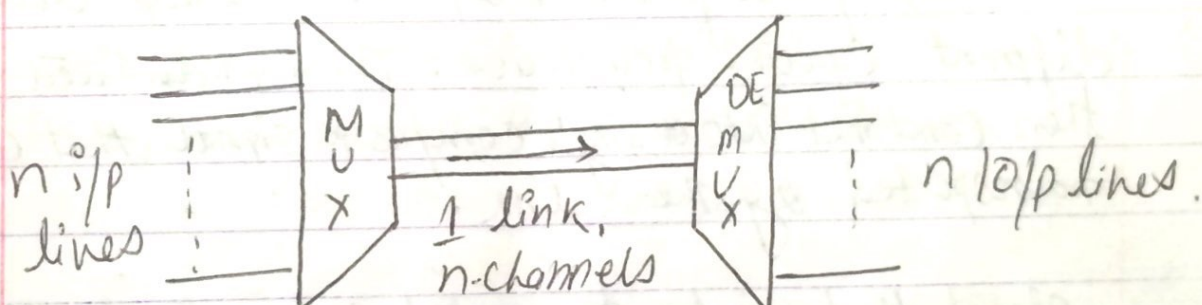
Multiplexing & switching

Swif

Multiplexing :-

It is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

- In a multiplexed system, n lines share the bandwidth of one link. The basic format of a multiplexed system is shown in figure.



- In above figure the ~~four~~ ^{'n'} device on the left direct their transmission streams to a multiplexer (mux) which combine them into a single stream.
- At receiving end that stream is fed into a de-multiplexer (DEMUX) which separates the stream back into its components transmission (one to many)

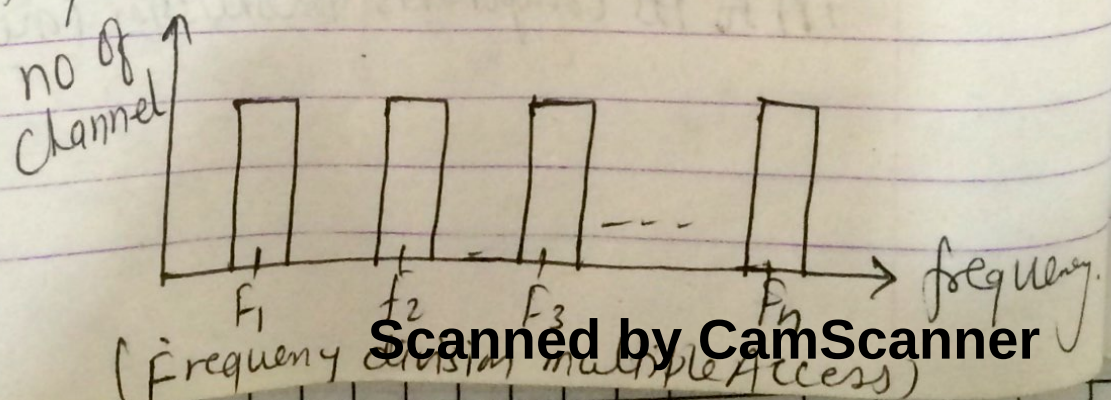
Categories of multiplexing:

- There are three basic multiplexing techniques.

- 1 Frequency Division Multiplexing (FDM)
- 2 Wavelength Division multiplexing (WDM)
- 3 Time division multiplexing (TDM).

① Frequency Division Multiplexing (FDM).

- In FDM, signals generated by each sending device, modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link.
- Signal that can be transported by sufficient bandwidth to accommodate the modulated signal.
- Channels can be separated by strips of unused b/w called guard bands to prevent signals from overlapping & prevent interference betⁿ carrier frequencies & data frequencies.

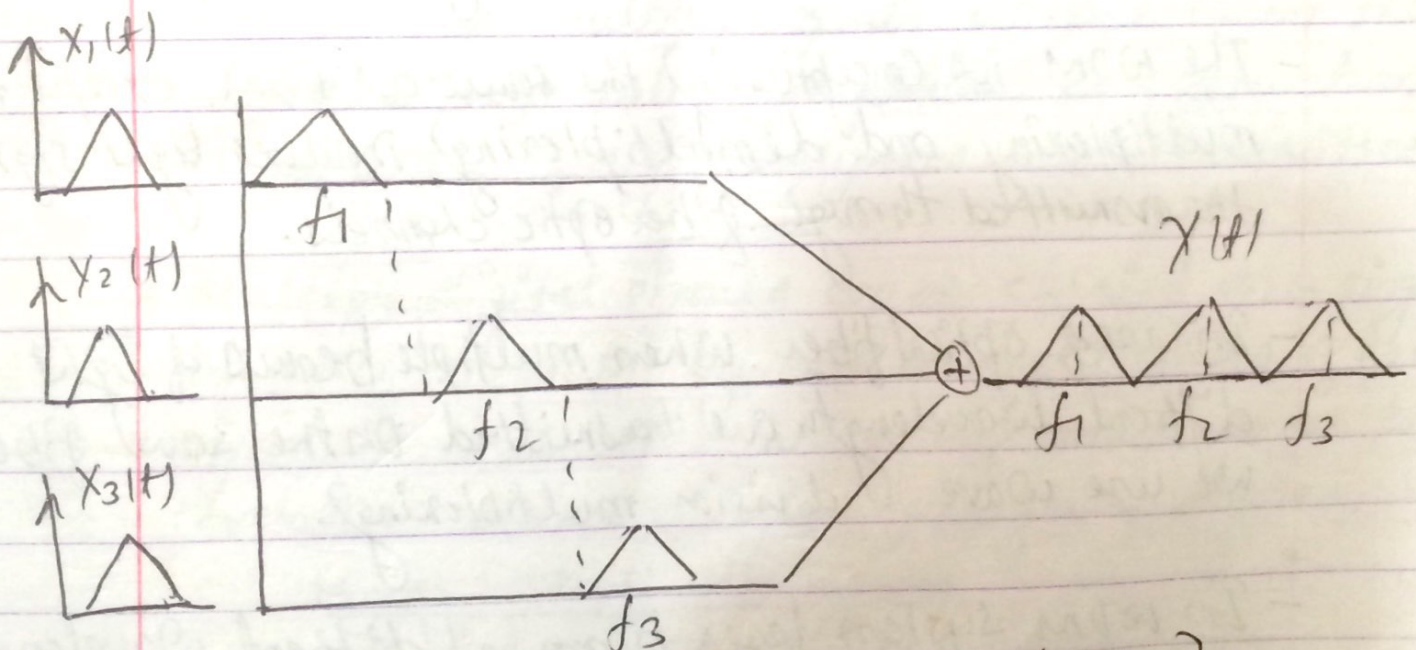


features

- Combination of multiple channels in freq domain.
- old & mature technology.
- Initially used in analog system.
- Simple ckt.
- No need to take care of timing?

Drawbacks.

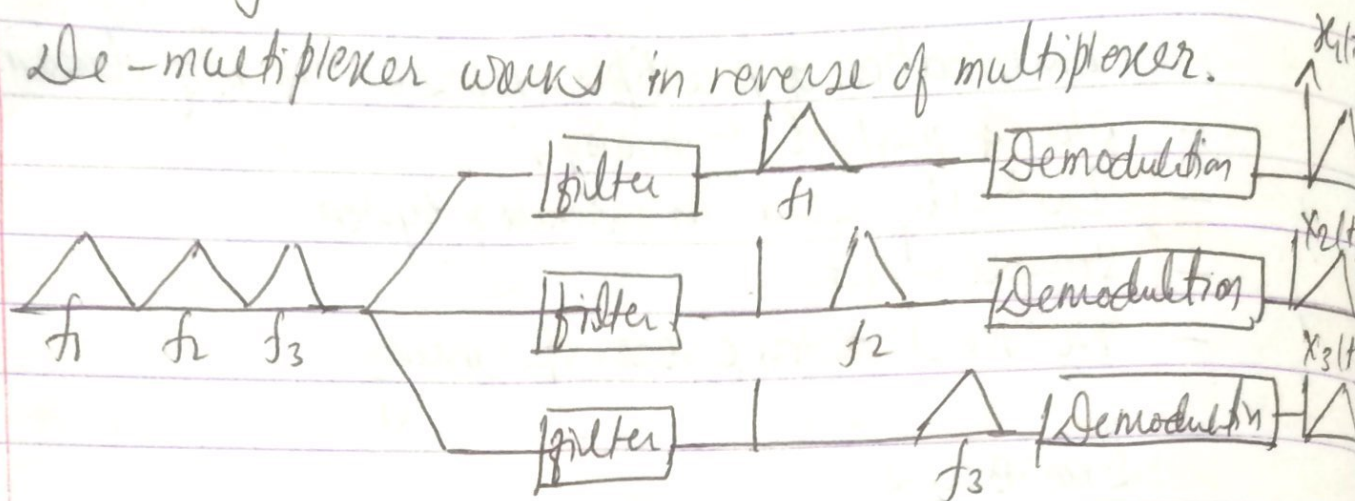
- Freq. resource is limited.
- unsecure
- spectrum management is difficult.
- Interference occurs bet. two channels.



(multiplexing in frequency domain)

above fig. illustrate the concept of multiplexer.

- De-multiplexer works in reverse of multiplexer.



(De-multiplexing in frequency domain).

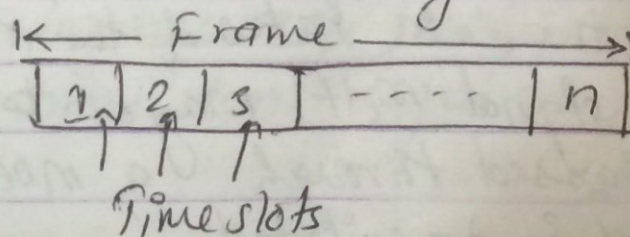
② Wavelength division Multiplexing (WDM).

- The WDM is conceptually the same as FDM, except that multiplexing and demultiplexing involves light signals, transmitted through fiber optic channels.
- In case of optical fiber when multiple beams of light of different wavelength are transmitted on the same fiber we use wave division multiplexing.
- In WDM system laser beam at different wavelengths generated by number of sources are sent to a multiplexer, which consolidates the sources for transmission over a single fiber line.

- After transmission of signal over long distance (low signal degrade & high data rate in optical fiber) the composite signal arrives at a demultiplexer, where the component channels are separated & sent to receivers at the destination point.
- Most WDM systems operate in the 1550nm range. In early systems 200nm was allocated to each channel, but today most WDM systems use 50nm spacing.

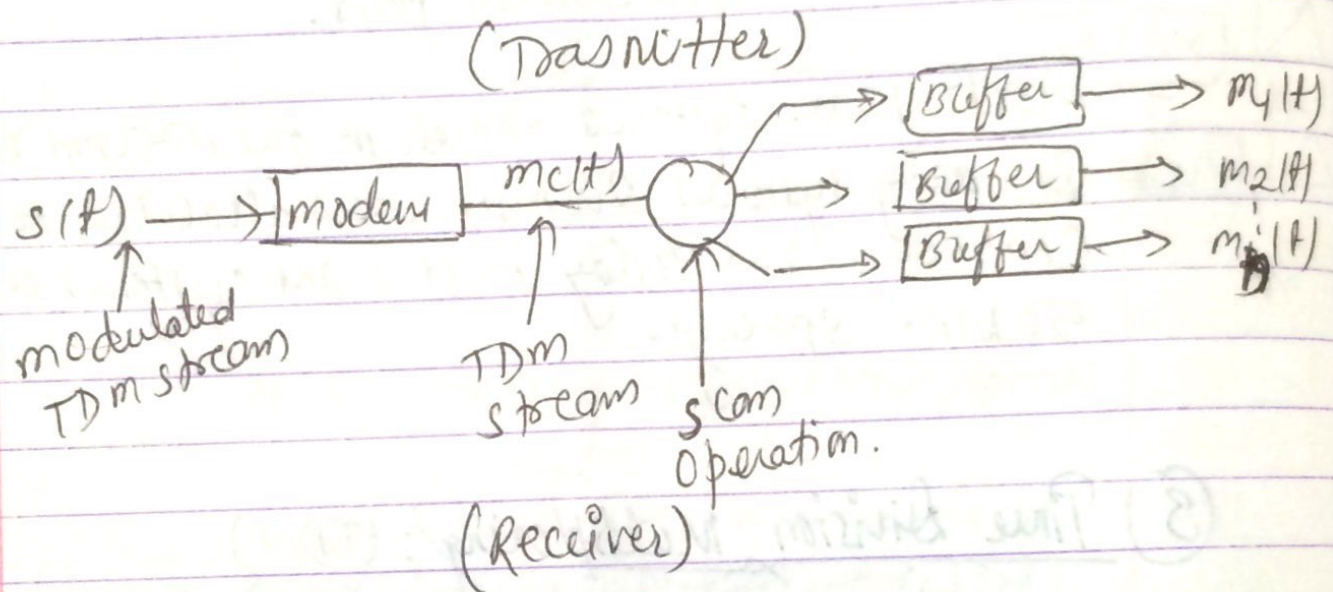
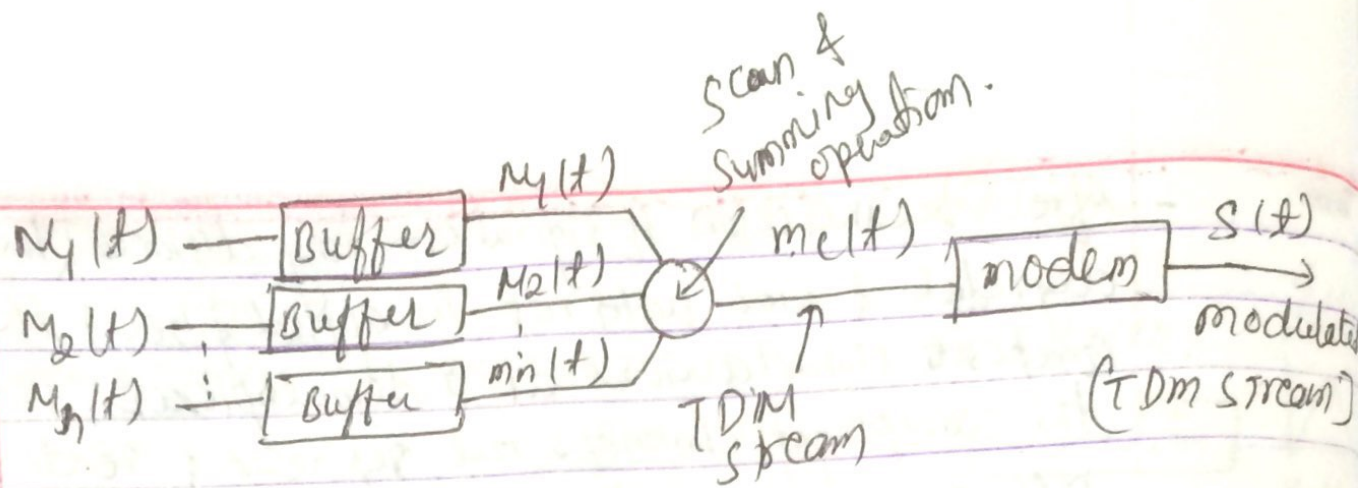
③ Time Division Multiplexing: (TDM)

- Time division multiplexing is possible when the achievable data rate of medium is greater than the data rate of digital signals to be transmitted.
- Multiple digital signals can be carried on a single transmission path by interleaving portions of each signal in time. The interleaving can be at the bit level or in blocks of bytes or larger quantities.



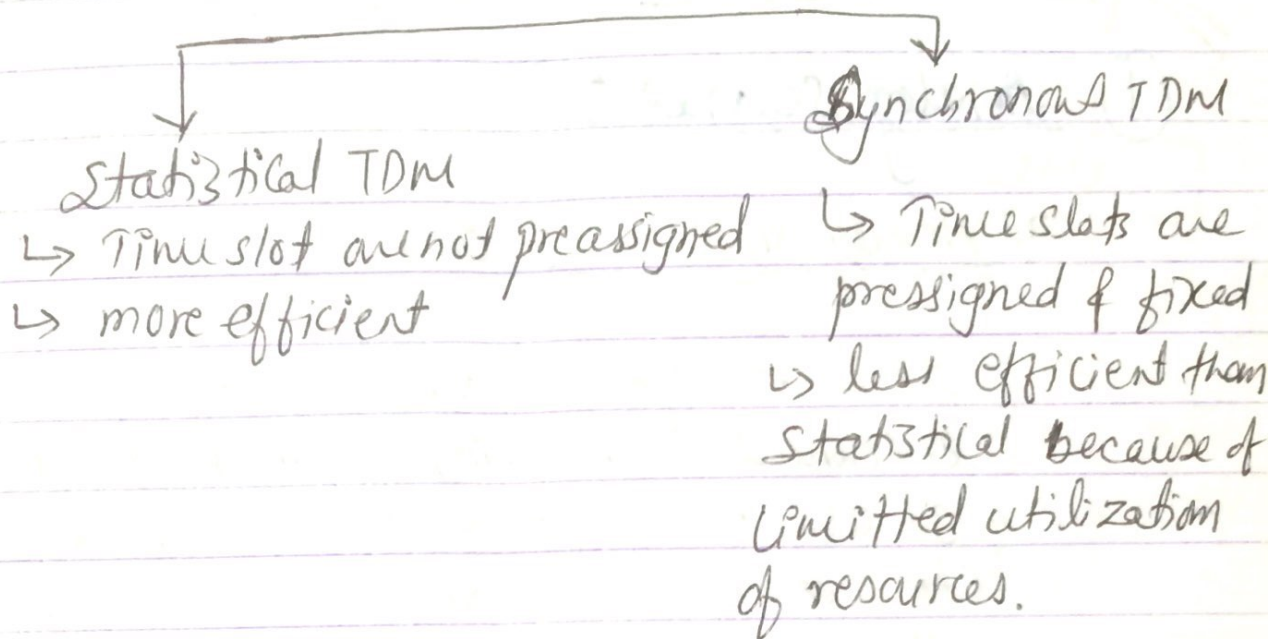
(may be empty or occupied.)

(fig. - TDM frame)



- The incoming data from each source are briefly buffered. Each buffer is typically one bit or one character in length.
- The buffers are scanned sequentially to form a composite digital data stream $m_c(t)$.
- The scan operation is sufficiently rapid so that each buffer is emptied before more data can arrive. Thus the data rate of $m_c(t)$ must at least equal the sum of the data rates of the $m_i(t)$.
- The digital signal $m_c(t)$ may be transmitted directly or passed through a modem so that an analog signal is transmitted.

TDM



Advantage of TDM over FDM.

- ① Digital technology (easy to implement)
- ② Secure "
- ③ Encryption possible
- ④ High capacity.

Drawbacks.

- complex
- Timing accuracy is important
- Synchronization required.

Telephone system

① Analog services :-

a) Analog switched services

- It is the familiar dialup service most often encountered when using a home telephone.
- It uses two or four wires i.e. twisted pair cable to connect the subscriber's handset to the n/w via an exchanger, this connection is called local loop.
- The n/w is sometimes referred to as a public switched telephone n/w (PSTN) with switched lines, when the caller dials up the conveyed to a switch, or series of switches at the exchange.
- The appropriate switches are then connected by activating according to the dial no.

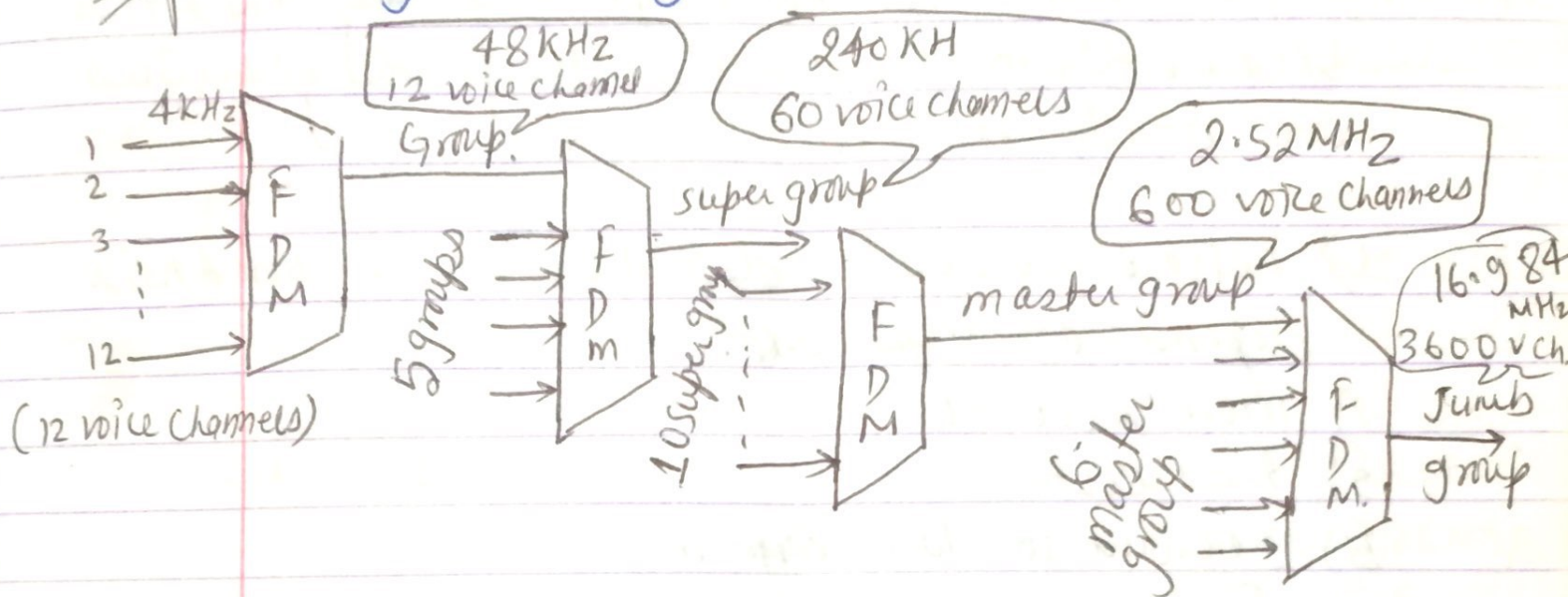
b) Analog Leased Services

- It offers customers further opportunity to a leased line sometimes called a dedicated line i.e. permanently connected to another customer.
- Although the connection still passes through the switches in the telephone n/w, subscribers experience it as a single line because the switch is always closed.

20/0, 20/11

Sup

Analog Hierarchy



- To maximize the efficiency of infrastructure the telephone companies have conditionally multiplexed signals from lower b/w lines onto higher b/w lines.
- In this way, many switched or leased lines can be combined into fewer but higher channels. For analog lines FDM is used.

Mechanism: -- In analog hierarchy, 12 voice channels are multiplexed onto a higher b/w line to create a group which has 48 KHz of b/w & supports 12 voice channels.

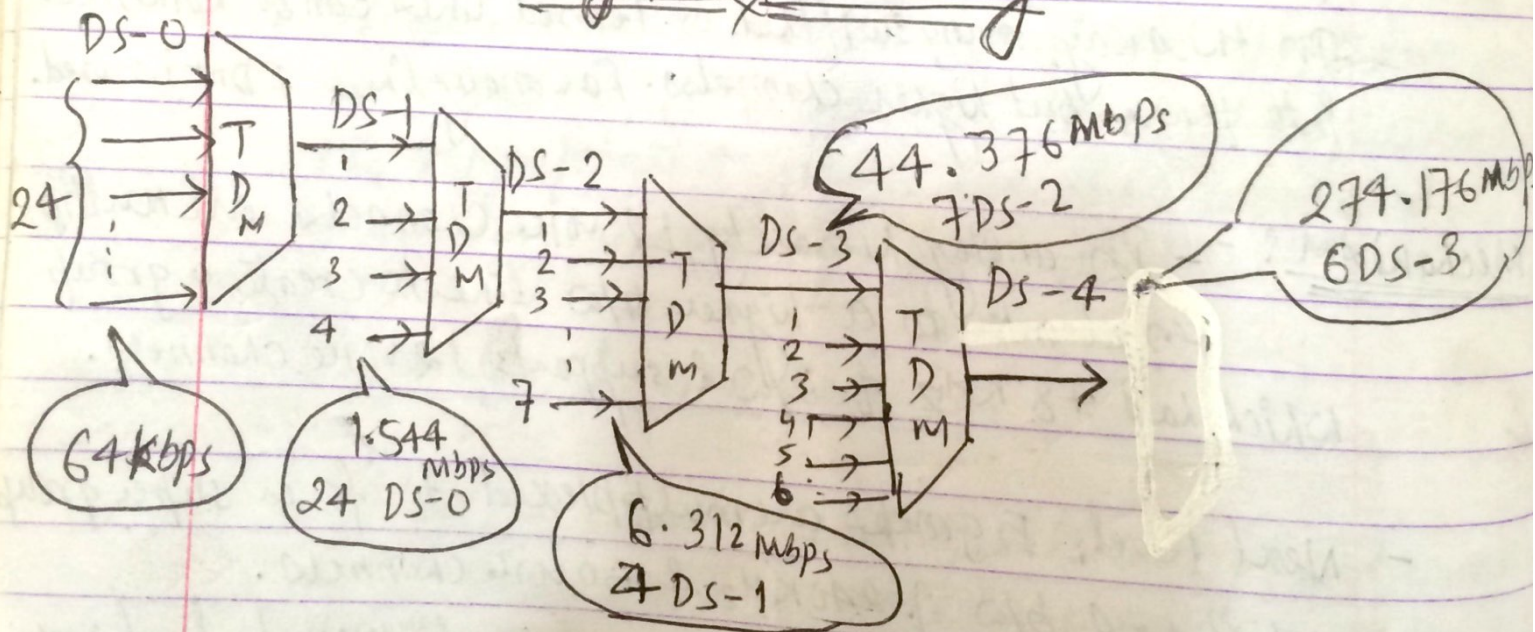
- Next level, 5 groups are multiplexed to form super group which has b/w of 240 KHz & 60 voice channels.
- At next level 10 super group are multiplexed to form master group having b/w 2.52 MHz ($2.4 \text{ MHz} + 0.12 \text{ MHz}$ guard band) & support 600 voice channel.
- Finally 6 master are multiplexed to form one **3600 voice channel master group** which has 16.984 MHz & support 3600 voice channels.

② Digital Services.

Digital services are most popular during this time because digital services have the following services over analog services.

- 1) noise is immune in digital service because noise depends on amplitude.
- 2) more Accurate.
- 3) less cost
- 4) secured due to encryption
- 5) Fast.

One of the best digital services is digital signal service or Digital Hierarchy



(fig:- Digital Hierarchy)

- It uses TDM in digital hierarchy

DS - Digital signal

Mechanism

- A DS-0 service is a single digital channel of 64 kbps. Where 24 DS-0 are multiplexed into a single DS-1 having 1.54 mbps data rate.
- At next level 4 DS-1 are multiplexed into a single DS-2 having data rate ~~6.312~~ 6.312 mbps.
- Finally At next level 7 DS-2 are multiplexed into a single DS-3 having data rate 44.376 mbps.
- Finally 6 DS-3 are multiplexed into a single DS-4 having data rate 274.176 mbps.

Setup

Switching:-

- switching is a solution of point-to-point connection (costly & inefficient).
- A switched n/w consists of a series of interlinked nodes called switches.
- switches are hardware & /or s/w device capable of creating temporary connection betⁿ two or more device.
- Traditionally there are 3 methods of switching

2006 Q. What is packet switching? How the i/p crt. can be multiplexed in analog hierarchy.

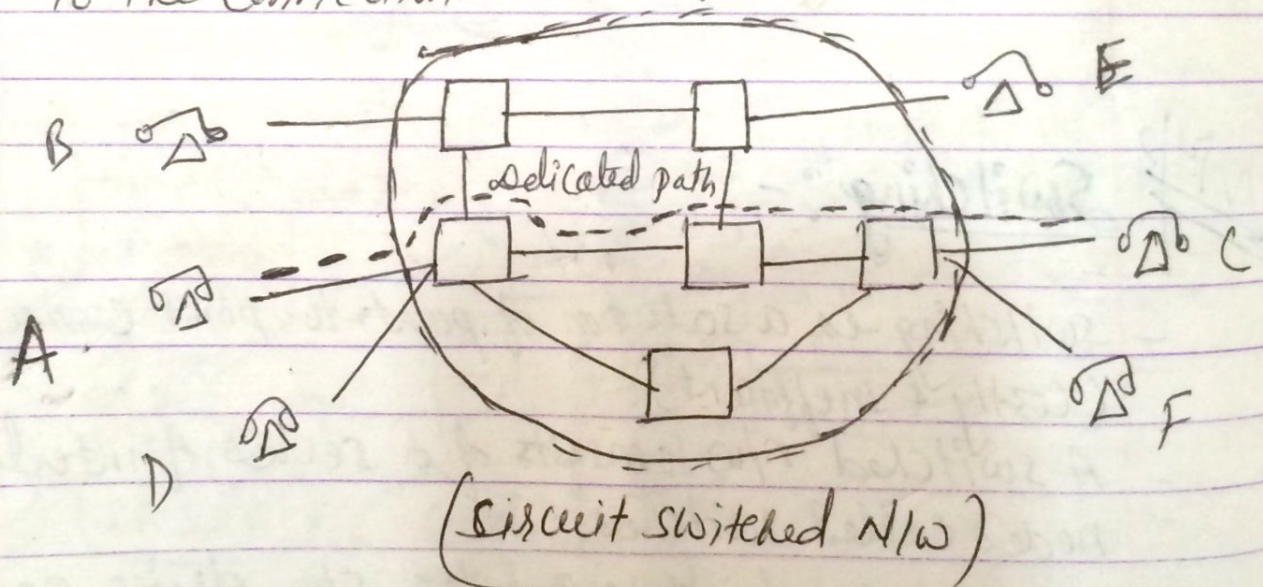
- (1) Circuit switching
- (2) Packet "
- (3) Message "

New switching

- ATM
- Frame Relay.

(1) Circuit switching :-

- This is a dedicated comm. path betⁿ two stations.
- on each physical link, a logical channel is dedicated to the connection.



Circuit switching involves three passes.

- (1) Circuit establishment : Before any signals can be transmitted, an end to end crt. must be established.

20/12/2010
Q. Explain the term message, packet, & ckt. switching.

- (i) Data Transfer: Information is transmitted from source to destination (eg A to C in figure) through the M/W
- (iii) Circuit disconnect:- After some period of data transfer, the connection is terminated.

application

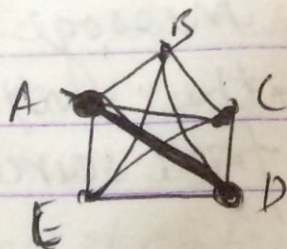
- (i) public Telephone N/W. (PSTN)
- (ii) Private Branch exchange (PBX)

Drawbacks

- In ckt. switching, the connection path is established before data transmission begins.
- Thus, Channel Capacity must be reserved betⁿ each pair of nodes in the path & each node must ~~be must~~ have internal switching capacity to handle the request connection.

But

- It is faster than other switching technique.

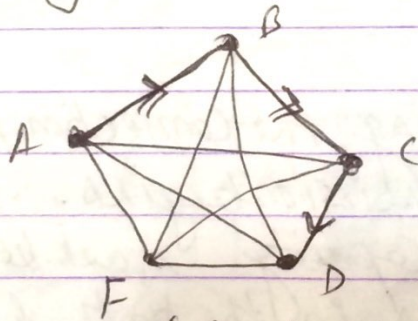


A - D

(SW) 2013

② Message switching

- store & forward concept.
- Each node has storage device, processor & commⁿ system.
- storage device store the message.
- processor process according to message & determine the destination of message.
- commⁿ system forward the message to correct destination.
- In this switching path sharing may occur.
- due to storing node to node it is slowest process.



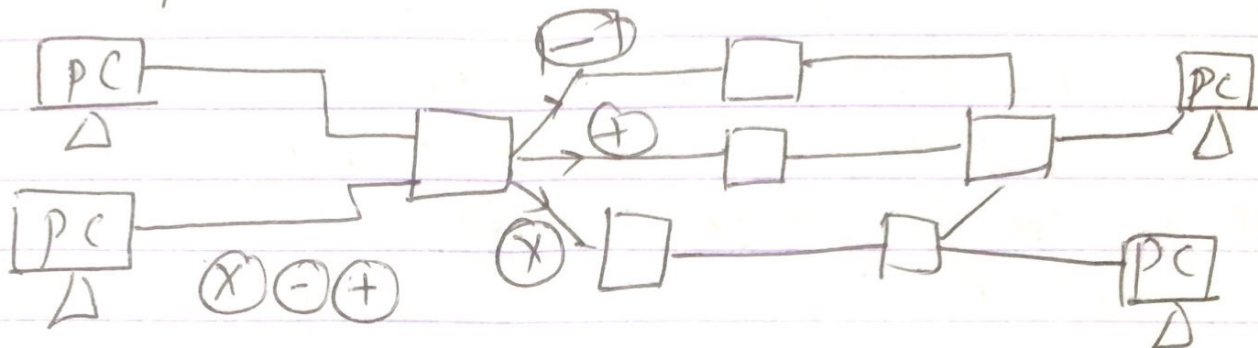
(A-D message switching)

③ Packet switching

(SW) 2007

- In packet switching, each message is broken down into fixed length message called packets.
- These packets travel in the N/W independent of the other packets from source to destination.

- however a mechanism is needed to track these individual packets & assemble them in order upon reception of the destination.



(packet switching n/w)

Packet switching approach.

① Datagram approach

- Each packet is treated independently with no reference to packets that have gone before
- Each packet contains complete control information & data.

② Virtual approach ckt.

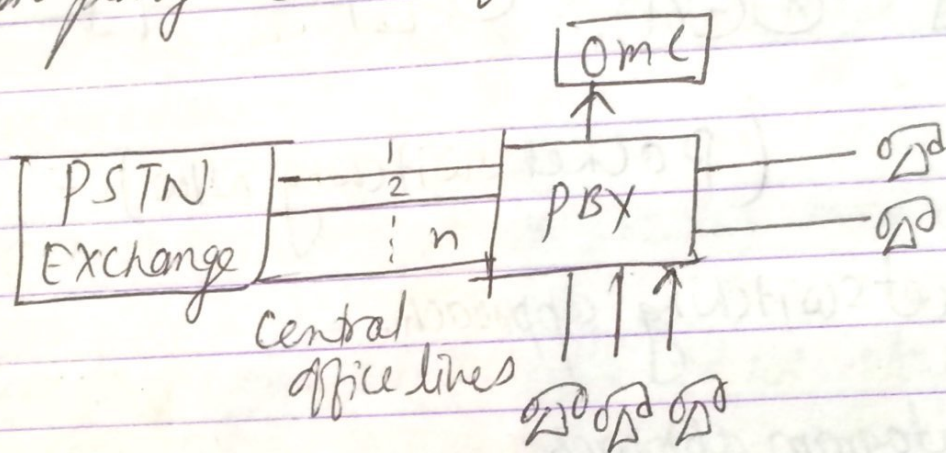
- Here a preplanned route is established before any packets are sent.
- once the route established, all packets should follow this same route through ~~this same route~~ the n/w betⁿ stations.
- Each packet contains a virtual ckt. identifier as well as data.

(SW)

2006

Private Branch Exchange (PBX)

- It is used to interconnect telephones within a building or office.
- The main purpose of a PBX is to save the cost of requiring a line for each user to the telephone company's central office.



(fig:- PBX)

PBX includes

- multiple phone lines (telephone trunk) that terminate at the PBX.
- OMC (Operation & maintenance Computer). A computer with memory that manages the switching of the calls within the PBX & in & out of it.
- The no. of lines within the PBX
- An operator console or switchedboard for a human operator.